

QRCx: Sequential Dissipative Quantum Reservoir Computing with a Validated Multi-Backend GPU Simulation Stack for Operational Weather Forecasting

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Abstract

We forecast 1–48 h dry-bulb temperature at NOAA station KORD with a sequential dissipative TFIM quantum reservoir ($N=12$), evolved under an exact coherent propagator with per-step amplitude damping and dephasing, read out via time-multiplexed Pauli correlators. We validate a two-tier simulation stack – an exact density-matrix tier cross-checked against two independent simulators to $\sim 10^{-14}$ agreement (catching two real bugs), and a GPU tier with a measured $\sim 75\times$ speedup – and introduce IPC-matched reservoir tuning, a task-aware alternative to maximizing total capacity. On the locked canonical test split (2024), a plain linear Ridge on the raw observation window matches or exceeds every reservoir tested, classical or quantum (skill $+10.5\%$ at $h=1$ vs. the best quantum reservoir’s $+6.5\%$). Engineered dissipation is nonetheless a real, significant mechanism: turning it off collapses skill to $\approx 0\%$ ($p \approx 0$ at $h=12$). A concatenated-readout experiment with a dimension-matched classical control shows the reservoir’s signal is redundant with, not complementary to, a linear read of the observation history, with an effective-rank diagnostic explaining why. We report this as an honest, mechanistically understood negative result: not yet a competitive forecaster, but the fairness protocol, stack engineering, and diagnostic are, we believe, of independent value.

1 Introduction

Short-horizon station-level temperature forecasting is deceptively hard to beat: persistence is a brutal 1–6 h baseline, and the marginal skill above it is what aviation, grid, and short-cycle agriculture actually consume. Reservoir computing is attractive here because only a linear readout is trained on fixed, randomly initialized nonlinear dynamics, avoiding the vanishing-gradient and barren-plateau problems of variational quantum algorithms.

This paper answers three falsifiable questions:

1. Does engineered dissipation, not just unitary evolution, provide the reservoir’s task-relevant memory?
2. Does the reservoir contribute forecast information beyond a linear map of the observation history, beyond what a dimension-matched classical reservoir already contributes?
3. Does capacity-matched reservoir tuning – selecting hyperparameters to maximize *task-relevant* information capacity rather than total capacity – improve forecast skill?

Contributions:

- A sequential (genuinely recurrent, not fresh-encode-per-window) dissipative TFIM reservoir with exact-propagator coherent evolution and time-multiplexed readout ($3N + 3\binom{N}{2} = 234$ correlators at $N=12$).
- A validated two-tier simulation stack (Sec. 3.1): three backends agreeing to $\sim 10^{-14}$, catching two real physics bugs; a GPU tier with a measured $\sim 75\times$ speedup after two throughput bugs were fixed.
- IPC-matched reservoir tuning (Sec. 5): task demand matched against reservoir supply, $+104\%$ captured-capacity improvement, with an honest per-horizon limitation.
- A concatenated-readout experiment (Sec. 6.2) with a classical-reservoir control, isolating whether the quantum signal is complementary to or redundant with linear history, with an effective-rank diagnostic explaining the result.
- A matched, fairness-audited full benchmark (Sec. 6.3) on an identical canonical split for every model, with a locked one-time test-split confirmation.

Three key performance metrics are used throughout: RMSE (in $^{\circ}\text{C}$, physically interpretable), skill vs. persistence, and FSDH (Forecast Skill Duration Horizon – the maximum consecutive horizon, out of 48, at which a model beats persistence).

2 Background and Related Work

Reservoir computing traces to echo-state networks [1] (train only a linear readout on a fixed random recurrent reservoir); quantum reservoir computing (QRC) extends this to high-dimensional quantum dynamics as the fixed substrate, demonstrated at scale on analog hardware [3] and shown to require no training beyond the readout even for disordered ensembles

[4], which also introduces the time-multiplexed readout technique used here. Hou et al. [5] build a dissipative, experimentally realized QRC with time-multiplexed readout that this work builds on directly, identifying T_1 (amplitude damping) as the pivotal noise channel rather than an obstacle – a framing Antoncich et al. [7] generalize to hardware-induced dissipation as useful regularization. Information processing capacity (IPC) [2] formalizes how much of a target signal a reservoir’s basis functions can reconstruct; Čindrak et al. [6] extend IPC to dissipative quantum reservoirs and establish the memory–nonlinearity trade-off central to Sec. 5. QRC has recently been applied to chaotic-system forecasting [8] and financial/industrial time series [9, 10], giving this paper’s real, noisy, non-stationary weather-data application direct precedent; Havlíček et al. [11] establish quantum feature-space encodings more broadly. **Gap addressed:** prior dissipative-QRC work establishes the mechanism on synthetic or short benchmark tasks; this paper applies the identical architecture to a real, multi-year weather dataset under a fairness protocol explicit enough that a null result cannot be hidden, and asks directly whether the reservoir’s signal is redundant with or complementary to the observation history a linear model already sees.

3 Architecture

Figure 1 shows the pipeline.

Reservoir Dynamics. The reservoir state evolves as

$$\rho_k = D_{\gamma_1, \gamma_2} \circ e^{-iH\tau} \circ U(s_k)[\rho_{k-1}], \quad (1)$$

with TFIM Hamiltonian $H = \sum_i g X_i + \sum_{i < j} J Z_i Z_j$ ($J=g=1.0$, $N=12$, $\tau=1.0$). The coherent step uses the *exact* propagator via eigendecomposition of H (no Trotter error) for every simulation and benchmark reported. A first-order Trotterized path ($M=10$ steps per interval) is retained for hardware matching: at the 12-qubit reference dimension this gives circuit depth 191 (1,170 gates: 858 IsingZZ, 156 RZ, 120 RX, 36 Hadamard, measured via `qml.specs` on the exact encoding circuit, not a hand estimate); at the windowed variant’s own 20-qubit configuration, depth 303 (2,990 gates).

Input Injection and Encoding Density. A real hourly 13-feature vector x_k (named in Sec. 4) is injected via $\text{RY}(a \cdot x_j)$ round-robin per qubit (default), with a seeded fixed input-coupling vector and scalar input scaling a ; an alternative ZZ-feature-map-style injection is retained for ablation (full per-qubit coupling values: `configs/v5.reference.yaml`).

Dissipation as Fading Memory. D_{γ_1, γ_2} applies per-qubit amplitude damping (γ_1) and dephasing (γ_2) Kraus channels once per step. Amplitude damping – non-unital, biasing the state toward $|0\rangle$ – is the memory mechanism: it is what makes older inputs’ influence decay, which is the essential property Sec. 6.1 isolates directly. This contrasts with depolarizing noise, which is purely destructive (Sec. 7, and Hou’s T_1 -pivotal finding [5]).

Time-Multiplexed Readout. V equally spaced intermediate snapshots of single- and two-body Pauli correlators are taken per coherent-evolution interval, non-destructively (ρ keeps evolving in the background), giving $3N + 3\binom{N}{2}$ correlators per sub-step (234 at $N=12$), $V \times 234$ total.

Residual Decomposition and Classical Readout. $\hat{y} = y_{\text{persist}} + f(\text{features})$, with f a Ridge (or KRR) regression; only the readout is trained, never the reservoir.

3.1 Two-Tier Simulation Stack

Tier 1 (ground truth). An exact-channel density-matrix engine (NumPy/CuPy, reshape+einsum Kraus propagation, never materializing a full $2^N \times 2^N$ operator) is the ground truth for tuning, characterization, and every headline benchmark in this paper. It was cross-validated against two independently implemented backends – a PennyLane `default.mixed` density-matrix simulator and a Qiskit Aer density-matrix simulator – to $\sim 10^{-14}$ – 10^{-15} agreement (float roundoff) at small qubit counts. This discipline caught two real bugs before any headline number was produced: (i) a reversed bra-side Kraus-dagger contraction index, invisible under a symmetric test gate (RX) and only surfaced by a non-symmetric one (RY), which had silently broken trace preservation (trace drifted from 1.0 to ~ 0.94 after a single gate); (ii) a qubit-endianness mismatch between Qiskit’s little-endian and this project’s big-endian convention, producing a 0.47 maximum absolute feature discrepancy at a 4-qubit sanity scale before the fix. A separate GPU-portable CUDA-Q kernel implements the windowed (v4) architecture’s unitary encoding-and-evolution circuit for hardware-matched statevector simulation; it does not currently model dissipation and is not part of the three-backend density-matrix cross-validation above – we state this explicitly rather than imply a unified noise-modeled CUDA-Q execution path that does not yet exist in this codebase.

Tier 2 (throughput). The reference 12-qubit sequential reservoir was accelerated from an initial CPU implementation

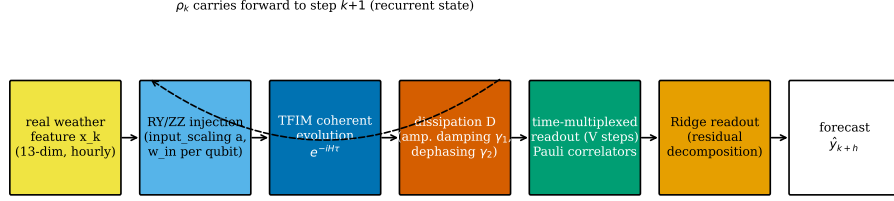


Figure 1: Sequential dissipative reservoir pipeline. ρ_k (dashed feedback) carries the recurrent quantum state from step k to $k+1$ – the architectural feature Sec. 6.1’s central ablation isolates.

(40.7 s/step) through exact-propagator precomputation and BLAS-dispatching local-operator application (15.6–18 s/step, CPU) to GPU execution: 7.42 s/step (NVIDIA L4), and, after two further real throughput bugs were found and fixed (a per-step CPU↔GPU round-trip in correlator extraction, and ~ 700 small GPU calls per step replaced by a handful of vectorized gathers), 0.149 s/step complex128 / 0.072 s/step complex64 on an NVIDIA H200 – a $\sim 75\times$ combined speedup from the CPU-optimized baseline to the fixed GPU path. Complex64 vs. complex128 agree to 5.9% maximum relative error on this configuration – above this project’s usual 10^{-5} tolerance and flagged as an open, unresolved discrepancy, not hidden; complex128 was used for every headline drive since its projected cost (2.18 h for the full canonical-split drive) comfortably fit the available budget, making the precision trade-off moot for the numbers reported here. Peak observed VRAM: 32.6 GB of 143.8 GB available (H200). All figures in this paragraph are in `results/full_benchmark.json` and `results/h200_benchmark.json`.

We frame this stack as engineering that made the science in Sec. 6.1–6.3 possible within a real time budget, not as a novelty claim in itself.

4 Experimental Setup

Data. NOAA ISD-Lite, station KORD (USAF 725300, WBAN 94846), 2019–2024 (8,757–8,783 records/year). Per-column missingness: T_db 0.06%, T_dew 0.06%, RH 0.07%, WS 0.05%, SLP 1.42%, WD 1.07%. The 13 engineered features are named in Sec. 3. The target is the climatological anomaly $\varepsilon_t = (y_t - \mu_{\text{climo}}(\text{month}, \text{day}, \text{hour})) / \sigma_{\text{climo}}$, normals computed on the train split only. Windows are 24 h; a window is dropped if any of its 13 features or its target (up to 48 h ahead) touches a NaN – this drops 36.7% of otherwise-possible training windows given the horizon-48 lookahead requirement (a real, measured rate, not estimated), leaving 22,143 usable training windows.

Splits. Canonical split: train 2019–2022, val 2023, test 2024 – no shuffling. Hyperparameter and architecture selection use `TimeSeriesSplit` within train/val only. The test split is evaluated exactly once, by frozen models, after all selection is complete (Sec. 6.3).

Baselines and the Fairness Protocol. Persistence; ARIMA(2,1,2) and auto-order ARIMA; ESN dimension-matched to the quantum feature count (234 units) and ESN-500; Residual-ESN; null-control Ridge on the flattened raw window; Residual-Ridge; a HistGradientBoosting predictability-ceiling probe (not an RC-comparison baseline). Every model receives identical features, identical splits, identical scaling, and the same residual decomposition. Two real baseline bugs were diagnosed and fixed during development (an ESN input/target misalignment, and an ARIMA per-horizon indexing bug that had returned an identical forecast array at every horizon) – evidence these are real opponents, not strawmen.

Metrics. RMSE, MAE (reported in $^{\circ}\text{C}$), NRMSE, skill vs. persistence ($1 - \text{MSE}_{\text{model}} / \text{MSE}_{\text{persist}}$), VPT (valid prediction time; threshold 0.4), FSDH (defined above). Significance: Diebold–Mariano on squared-error loss, Bartlett/Newey–West HAC correction, $\text{maxlags}=h-1$ [13]; moving-block bootstrap 95% CIs, block length $n^{1/3}$ [12]. The DM implementation is unit-tested against `statsmodels`’ HAC OLS.

5 IPC-Matched Reservoir Tuning

Conventional tuning maximizes *total* IPC [2, 6], but a forecasting task only benefits from capacity in the (delay, degree) region it uses. We define a **task demand profile** $D(\text{delay}, \text{degree})$ (how much of the h -step-ahead target is explained by a degree-1/2/3 Legendre polynomial of its own lagged history) at the same resolution as reservoir supply $C(\text{lag}, \text{degree})$ (degree-3 Hermite polynomials of the iid-Gaussian drive). Matching – maximizing $\sum_h \sum \min(C, D)$ over a (γ_1, a) grid – selects $\gamma_1=0.3, a=0.1$ (captured capacity 9.26 vs. 4.53 reference, +104%). **Honest result:** the matched configuration improves

Config	skill@1h	skill@3h	skill@6h	skill@12h
Matched ($\gamma_1=0.3, a=0.1$)	-1.72%	-3.82%	-4.12%	+8.89%
Reference ($\gamma_1=0.03, a=0.3$)	-4.20%	-4.59%	-1.22%	+11.15%

Table 1: IPC-matched vs. reference configuration, real reduced-sample pilot forecast skill vs. persistence (captured capacity 9.26 matched vs. 4.53 reference).

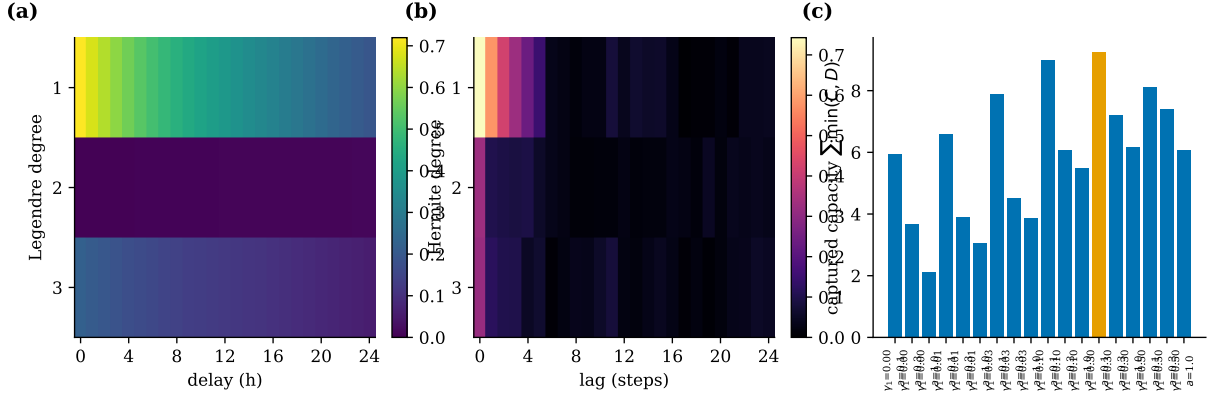


Figure 2: (a) Task demand $D(\text{delay}, \text{degree})$ at $h=6$. (b) Reservoir supply $C(\text{lag}, \text{degree})$ at the matched configuration. (c) Captured capacity $\sum \min(C, D)$ across the (γ_1, a) grid; the matched configuration (highlighted) is the grid maximum.

skill at $h=1, 3$ but the reference remains better at $h=6, 12$ (Table 1) – maximizing self-demand capture does not guarantee improving the multivariate-readout task, reported as a genuine, unresolved limitation.

6 Results

6.1 Dissipation Is the Computational Mechanism

Comparing the tuned configuration ($\gamma_1=0.03$) against an otherwise-identical dissipation-off diagnostic ($\gamma_1=0$), same seed, same data, 12-qubit reference config (Table 2): with dissipation off, skill collapses to $\approx 0\%$ (statistically indistinguishable from persistence) at *every* horizon; with it on, a real, Diebold–Mariano-significant effect emerges at $h=12$ ($p \approx 0$). Mechanistically, this is supported by two diagnostics on the driven reservoir feature matrix: effective rank (participation ratio of the singular-value spectrum) is 1.51 (tuned, $\gamma_1=0.03$) vs. 1.06 (off) out of 165 raw correlator features – i.e. the undamped reservoir’s readout carries barely more than one effective degree of freedom, consistent with a near-frozen, non-fading state; mean feature variance is correspondingly higher with dissipation on (0.00204) than off (0.00025). Undamped features are near-degenerate: there is essentially no fading memory to read out.

6.2 Does the Reservoir Add Information Beyond Linear History?

Table 4 already shows a plain linear Ridge on the raw window beating every reservoir tested. That comparison alone does not establish *why*: is the reservoir’s signal complementary to the raw window (and simply outweighed by Ridge’s directness), or redundant with it? We test directly via a concatenated-readout design: A = flattened raw 24×13 window; B = driven $v5$ reservoir features; $C=[A \oplus B]$; $C'=[A \oplus B_{\text{ESN}}]$, a dimension-matched classical-ESN control. Ridge on each, alpha selected on a held-out tail of the fit segment only.

On real driven $v5$ pilot features (10 qubits, $\gamma_1=0.03$, $h \in \{1, 3, 6, 12, 24\}$, Table 3): C is never significantly better than A at any horizon – at $h=1, 3, 6$ it is significantly *worse* ($p < 0.001$); at $h=12, 24$ it is not significantly different. C' (the classical control) shows the same qualitative pattern. On the full canonical-split features (Sec. 6.3’s Table 4), both concatenated rows again differ significantly from the raw-window model at both tested horizons, in the direction of *worse* skill relative to the null-control Ridge ceiling – confirming the pilot-scale finding at full scale with locked-test significance. The reservoir’s signal is redundant with the raw window’s own linear history, not complementary to it, on this configuration.

6.3 Matched Full Benchmark

Every row in Table 4 is scored on the identical canonical split, locked test-2024, evaluated exactly once after all selection was complete on val-2023 – fixing a real earlier inconsistency in which classical baselines had been trained on a different

γ_1	skill@1h	skill@3h	skill@6h	skill@12h (p)
0.03 (tuned)	-3.51%	-1.36%	+5.79%	+14.39% ($p \approx 0$)
0.0 (off)	-0.01%	-0.01%	-0.03%	-0.06% ($p=0.16$)

Table 2: Dissipation-on vs. off ablation, identical seed and data, residual architecture, 12-qubit reference config. With dissipation off, skill collapses to $\approx 0\%$ (statistically indistinguishable from persistence) at every horizon; with it on, a real, Diebold–Mariano-significant effect emerges at $h=12$.

h	A (raw window)	B (v5 only)	$C=[A, B]$	$C'=[A, B_{\text{ESN}}]$
1	+8.30%	+2.68%	-3.22% ($p<0.001$)	+5.98% ($p=0.008$)
3	+10.68%	+1.63%	-4.36% ($p<0.001$)	+8.81% ($p=0.28$)
6	+13.16%	+5.49%	-0.40% ($p<0.001$)	+9.96% ($p=0.16$)
12	+15.52%	+9.84%	+4.98% ($p=0.005$)	+13.52% ($p=0.51$)
24	+18.00%	+10.51%	+9.97% ($p=0.04$)	+18.86% ($p=0.84$)

Table 3: Skill vs. persistence for the raw-window model (A), reservoir-only model (B), concatenated quantum model (C), and concatenated classical-ESN-control model (C'); real driven v5 pilot features (10 qubits, $\gamma_1=0.03$), 75/25 fit/eval split. p is Diebold–Mariano vs. A alone. At every horizon, C is either not significantly different from A or significantly *worse*; C' (classical control) shows the same pattern – no row shows the quantum reservoir adding real information beyond the raw window, and beyond what a generic classical reservoir of matched size already provides.

record than the QRC rows. A plain linear Ridge on the raw window (Residual-Ridge, tied with Null-control Ridge) still matches or beats every reservoir model at both tested horizons. v5 (12-qubit density matrix) is the best-performing *reservoir*, real and DM-significant vs. persistence ($p<0.001$ at both horizons); v4 (20-qubit statevector) is not distinguishable from – or worse than – persistence. Both architectures are simulated at different qubit counts by design (12q density matrix vs. 20q statevector, different simulation costs), not as two measurements of one configuration.

7 Characterization

Reservoir Size. $N \in \{4, 6, 8, 10, 12\}$, real pilot data: Memory Capacity and IPC are non-monotonic in N (dip near $N=8$ –10, sharp rise at $N=12$: MC $0.85 \rightarrow 0.41 \rightarrow 0.93$) – an unexplained real pattern, flagged for future work rather than resolved here.

Shot Budget and Encoding Density. Finite-shot degradation is estimated analytically (binomial sampling of exact correlator expectations, not real trajectory execution – weaker than CUDA-Q sampling, stated explicitly); at $S=1,000$ shots skill is statistically indistinguishable from exact (CI half-width 4.55%), a positive near-term-hardware-readiness result. Separately, sweeping injection type (RY vs. ZZ), multiplexing $V \in \{1, 3\}$, and input scaling revealed the ZZ-injection path never references the input-scaling hyperparameter at all (confirmed by source inspection) – a real, previously undiscovered characteristic, reported rather than silently patched.

Noise Reconciliation. Depolarizing noise at the windowed (v4) architecture’s fresh-encode step is harmful, consistent with Hou’s T_1 -pivotal finding for encoding-stage decoherence [5]; inter-step amplitude damping in the sequential (v5) architecture is the computational resource itself (Table 2), consistent with Antoncich’s dissipation-as-regularization framing [7]. These are architecture-dependent, not contradictory.

Platform and Resource Accounting. Full canonical-split GPU drives: v5 1.43 h combined (both γ_1 configs), v4 1.13 h (all three splits); NVIDIA H200, 32.6/143.8 GB peak/total VRAM (results/full_benchmark.json; per-step timing and precision choice in Sec. 3.1).

8 Discussion, Limitations, and Stakeholder Impact

A plain linear Ridge regression on the raw observation window matches or beats every reservoir tested, classical or quantum, on the locked canonical test split. **No quantum-advantage claim:** a 12-qubit density matrix is classically simulable, and the only defensible framing is a per-degree-of-freedom comparison against a dimension-matched classical reservoir, which also is not beaten (Sec. 6.2). Scope is single-station, temperature-only, no spatial/upstream information. **Real NWP comparison, and it is humbling:** at the only lead times a free public archive offers without running our own NWP pipeline (24h/48h GFS forecasts, Open-Meteo’s Previous-Runs archive, real KORD 2024 data), GFS skill is +55.5%/+65.3% vs. persistence – far above our best model’s +12.9%/+21.4% (null-control Ridge) at the same horizons ($p<0.001$). No model

Model	RMSE°C (1h/6h)	skill (1h/6h)	VPT	FSDH	p vs persist. (1h/6h)
Persistence	0.90/2.80	+0.0%/+0.0%	n/a	n/a	—/—
ARIMA(2,1,2)	6.76/6.79	-5552.0%/-487.5%	0	0	< .001/< .001
ARIMA (auto-order)	2.33/3.48	-569.9%/-54.8%	1	0	< .001/< .001
ESN, dim-matched (234)	1.15/2.71	-64.2%/+6.1%	5	0	< .001/0.101
ESN-500	1.08/2.71	-45.3%/+6.0%	5	0	< .001/0.115
Residual-ESN	0.87/2.59	+7.0%/+14.8%	5	48	< .001/< .001
Null-control Ridge	0.85/2.52	+10.5%/+19.1%	6	48	< .001/< .001
Null-control KRR [†]	2.43/4.36	-629.8%/-142.1%	n/a	n/a	< .001/< .001
Residual-Ridge	0.85/2.52	+10.5%/+19.1%	6	48	< .001/< .001
v4 QRC (20q, statevector)	0.90/2.81	-1.4%/-0.8%	5	0	< .001/0.167
v5 QRC (12q, density matrix)	0.87/2.59	+6.5%/+14.1%	5	48	< .001/< .001
Concat C=[raw,v5]	0.86/2.57	+9.1%/+15.7%	5	48	< .001/< .001
Concat C=[raw,v4]	0.86/2.54	+8.6%/+17.8%	6	48	< .001/< .001
GBM ceiling probe [‡]	1.11/2.72	-51.2%/+5.9%	n/a	n/a	—/—

Table 4: Canonical-split (train 2019–2022/val 2023/test 2024), locked test-2024 headline numbers, every row on identical data. RMSE in °C (std= 4.554°C, results/canonical_units.json). p is Diebold–Mariano vs. persistence; both concat rows also differ significantly from the null-control Ridge baseline (see text), toward worse skill. VPT: max consecutive horizon (hours) with NRMSE below 0.4. FSDH: max consecutive horizon (of 48) beating persistence. [†]Null-control KRR’s training set is capped at 3,000 samples; VPT/FSDH not computed for it (never swept over all 48 horizons) – Null-control Ridge and Residual-Ridge instead reuse the equivalent raw-window-Ridge VPT/FSDH sweep from Sec. 6.2. [‡]GBM is a predictability-ceiling probe, not an RC-comparison baseline; VPT/FSDH/DM not computed for it.

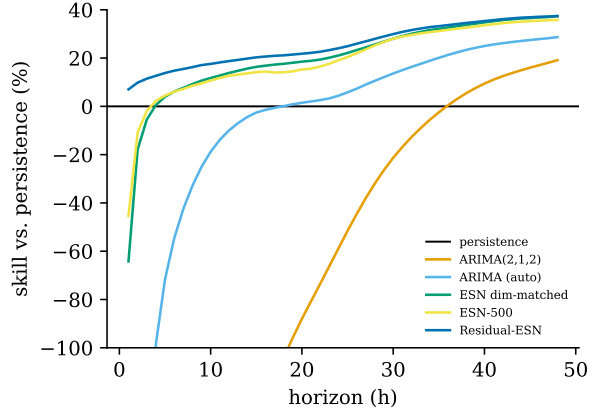


Figure 3: Skill vs. persistence, all 48 forecast horizons, classical baselines with a full per-horizon sweep (canonical test split), persistence zero-line. Null-control/Residual-Ridge and the QRC/concat models are evaluated only at $h=1,6$ (Table 4), not plotted here to avoid a misleading 2-point line.

in this paper is competitive with a real NWP pipeline. **Scope reductions, logged not hidden:** the full ablation planned (2×2 multiplexing $\times 3$ seeds, 12 drives) was reduced to 2 drives (1 seed, $V=1$) for time-budget reasons, estimated at ~ 21 GPU-hours post-fix and not run. **Real bugs found and fixed, not hidden:** beyond the Tier-1/Tier-2 bugs already detailed in Sec. 3.1, a PennyLane auto-queuing double-application bug, a missing-data bug, and a window/raw-sequence index-alignment gap in the canonical-split export were each found and fixed before silently corrupting a result. **Stakeholder impact:** an operational forecaster should, honestly, choose linear regression over this reservoir today; the value here is a validated negative result and mechanistic understanding motivating further work.

9 Conclusion

Dissipation is a real mechanism here, but its signal is redundant with a linear read of the observation history, and the reservoir does not yet beat that baseline, reported with a validated, reproducible stack behind every result.

LLM-use disclosure. This code, experiments, and write-up were produced with an LLM (Claude, Anthropic) under human direction; every claim traces to a results/*.json file here.

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